Priyanshu Upadhyay

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Education

KIET Group of Institutions — **Delhi NCR**
B.Tech in Computer Science and Engineering; CGPA: 7.5 — *2023 – Present*

Kanha Makhan Public School — **Mathura, UP**
Senior Secondary (Class XII) — CBSE, 90% — *2022 – 2023*

Projects

Customer Retention Intelligence | *FastAPI, XGBoost, Scikit-learn, Docker* | GitHub | Demo — **Mar – Apr 2026**

- Engineered a thread-safe Scikit-learn pipeline with custom sequential preprocessors and a parallel ColumnTransformer (StandardScaler + OneHotEncoder) feeding XGBoost — achieving 85% recall and 0.84 ROC-AUC.
- Designed a Recall@20% budget-ranking strategy capturing ˜50% of actual churners using only 20% of the retention budget, replacing naive threshold-based classification.
- Architected a decoupled FastAPI inference engine with async PostgreSQL telemetry logging (BackgroundTasks + asyncpg), then containerized it into a multi-service Docker Compose stack and deployed it on AWS EC2 — taking the system from local development to a production-hosted service with zero added UI latency.

QnA With Jupyter (Relational RAG System) | *LangChain, ChromaDB, Groq, Ollama, AST* | GitHub — **Feb – Mar 2026**

- Engineered a "Bait & Switch" retrieval pipeline: semantic search identifies a `cell_id`, then fetches the full uncut parent cell from a JSON ground-truth warehouse — eliminating context loss from naive chunking.
- Built dual-embedding ChromaDB pipelines — bge-m3 for NL semantics and CodeT5 for code syntax — and routed final Q&A through Groq API (Llama 3.3 70B) for low-latency inference; achieved ˜80% contextual accuracy.
- Deployed local Llama 3.1 (Ollama) with Python AST-based ingestion to parse notebook dependencies and auto-generate semantic cell descriptions — keeping sensitive data off cloud APIs while enabling rich indexing.

LoanClassinator (Explainable AI) | *XGBoost, SHAP, K-Means, LLM APIs* | GitHub | Demo — **Oct – Nov 2025**

- Built a dual-model engine on 2.2M records (150 features): XGBoost for default probability prediction alongside K-Means clustering for dynamic financial risk segmentation.
- Integrated SHAP (Explainable AI) to quantify top features per loan decision; fed risk scores, cluster stats, and SHAP values into a context-aware LLM agent generating personalised financial advice and alternate loan terms.
- Rejected SMOTE after A/B testing revealed calibration noise in high-probability prediction bounds; used native XGBoost class weights instead, improving reliability on the imbalanced 2.2M-record dataset.

Technical Skills

Languages: Python, C++, C

Machine Learning & Data Science: Scikit-learn, XGBoost, Pandas, NumPy, SHAP (Explainable AI), Feature Engineering

Generative AI & LLMs: LangChain, RAG Pipelines, LLM Integration, Prompt Engineering, ChromaDB (Vector DB)

Backend & Deployment: FastAPI, Docker, Docker Compose, AWS EC2, REST APIs

Databases & CS Fundamentals: PostgreSQL, SQL, OOP, OS, DBMS

Tools: Git, Streamlit, Jupyter Notebook, VS Code, OpenCV

Key Achievements

- **Clash of Codes** — Finalist, Top 1% (1000+ participants, 256+ teams): built an AI virtual canvas enabling air-drawing via hand gesture recognition, with voice-controlled navigation and Gemini integration to solve visually drawn problems.
- **HackSpire, IPEC Ghaziabad** — Finalist: developed a computer vision-based water footprint calculator that identifies crop types from images and estimates agricultural water consumption to guide sustainable irrigation decisions.

Position of Responsibility

Machine Learning Lead — **May 2025 – May 2026**
Technocrats KIET — *Delhi NCR*

- Guided second-year coordinators running ML/AI bootcamps for 100+ students, mentored 4+ students on project architectures across ML, CV, and GenAI, and resolved escalated technical issues including Jupyter Notebook environment setups.

Coordinator — **Oct 2024 – May 2025**
Technocrats KIET — *Delhi NCR*

- Organised regular ML workshops and a 3-day Computer Vision bootcamp — drew 60+ students, with 30 completing the full program and 90% independently building and running the final project.
- Taught ML fundamentals from Python basics to Linear Regression in workshops, and covered computer vision concepts culminating in an Automatic Number Plate Recognition (ANPR) project in the bootcamp.

Certifications

- **GitHub Copilot Certification** — Microsoft *[Verify]* — Jul 2026
- **AWS Certified Cloud Practitioner** *[Verify]* — Feb 2026
- **AWS Certified AI Practitioner** *[Verify]* — Feb 2026
- **Linear Algebra for Machine Learning and Data Science** — DeepLearning.AI *[Verify]* — Jan 2025